**CABINET MÉDICAL DES TILLEULS**
8 allée des Tilleuls, 71250 Bellerive-sur-Saône
Tél. 03 00 00 00 00

Docteur Claire VALMONT
Médecine générale

Bellerive, le 12/10/2026
Madame Inès MOREAU

# ORDONNANCE

Paracétamol 1 g — un comprimé trois fois par jour *pendant cinq jours*

Sérum physiologique — lavage nasal matin et soir

Renouvellement : non

CABINET MÉDICAL DES TILLEULS · BELLERIVE
Dr C. VALMONT
Médecine générale
71250 BELLERIVE

Signature du médecin

**MARCHÉ DU PONT**
Épicerie fine · Lugano
Via del Ponte 3

| | |
|---|---|
| Pane integrale | 3.40 |
| Formaggio Alpe | 7.95 |
| Olive verdi | 4.20 |
| Acqua frizzante 6x | 5.10 |
| Caffè in grani | 11.80 |
| **TOTALE CHF** | **32.45** |
| Carta | 32.45 |

14.10.2026 18:42 · Cassa 2
Grazie e arrivederci

4 006381 333931

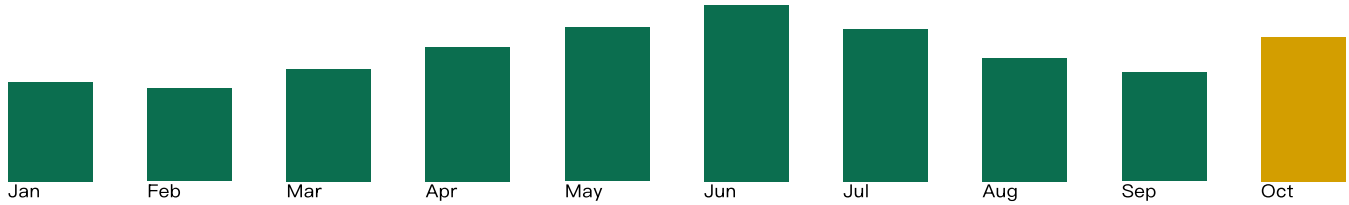
# GreenGrid Energy

Electricity & gas statement

Account 5503-8841-02
Statement date 5 Nov 2026
Billing period 1 Oct – 31 Oct 2026

| Previous balance | Payments received | New charges | Total amount due |
|---|---|---|---|
| $118.40 | -$118.40 | $142.73 | $142.73 |
| | | | Due 26 Nov 2026 |

## Your usage this year




## Charges

| Service | Usage | Rate | Amount |
|---|---|---|---|
| Electricity – first 300 kWh | 300 kWh | $0.1180 | $35.40 |
| Electricity – over 300 kWh | 186 kWh | $0.1420 | $26.41 |
| Natural gas | 41 therms | $0.9650 | $39.57 |
| Service & delivery charge | 1 month | $24.60 | $24.60 |
| Taxes and surcharges | | | $16.75 |
| **Total new charges** | | | **$142.73** |

Meter readings: electricity 48,310 → 48,796 · gas 3,402 → 3,443. Questions? Call 1-800-555-0142 or visit greengrid.example.

**Payment stub**
Account 5503-8841-02
Amount due $142.73 by 26 Nov 2026

5503884102 0014273 112626

GreenGrid Energy
PO Box 4410
Springfield, ST 00012

**SKYWAY AIRLINES · BOARDING PASS** — **ECONOMY**

PASSENGER
**LAURENT / CAMILLE MS**

FLIGHT
**SW 1827**

DATE
**14 NOV 2026**

FROM
# PARIS CDG

TO
# LISBON LIS

GATE
# K42

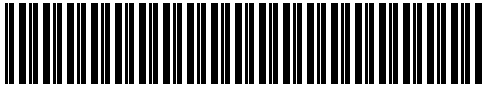
SEAT
# 17C

BOARDING
**09:05**

DEPARTS
**09:40**

GROUP
**B**

SEQUENCE
**112**



---

**SKYWAY AIRLINES**

PASSENGER
LAURENT / CAMILLE

FLIGHT · DATE
SW 1827 · 14 NOV

FROM → TO
CDG → LIS

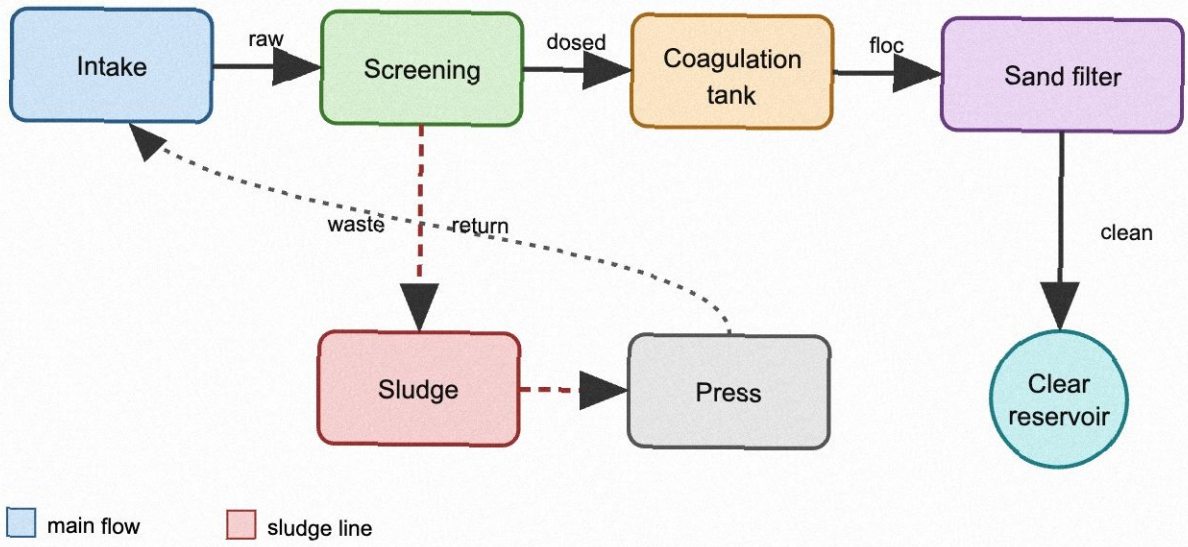
SEAT
**17C**



## 2. Water treatment process

The plant draws raw water from the river intake and passes it through four stages before it reaches the distribution network. Each stage is monitored separately, and samples are taken after every step so that a fault can be traced back to the stage where it started.

The diagram below shows the main flow and the return line used for sludge. Operators switch the bypass valve only during maintenance, when one of the filter beds is out of service.

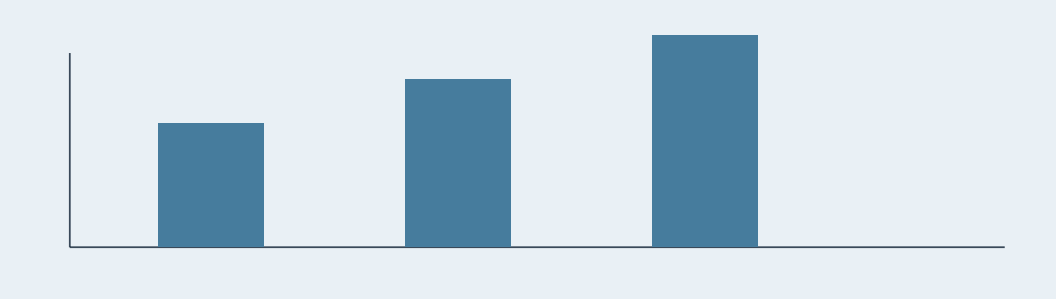



Sludge removed during screening is pressed and dried on site. The filtrate from the press is returned to the intake, which keeps the plant's losses below two percent of the treated volume.

Each filter bed is backwashed once a day. The clear reservoir holds enough water for twelve hours of normal demand, which leaves time to repair a fault without cutting supply.

# Quarterly review

Elena Martin | ID: DOC-113-1 | 2026-10-15



## Highlights

- Revenue grew by 12.5%.
- 128 projects completed.

preprint:2610.04417v2 [stat.ME] 9 Oct 2026

# Seasonal Patterns in Small-River Discharge

Lea Marchetti · Jonas Feld · Ana Ribeiro

## Abstract

We describe the seasonal behaviour of discharge in eleven small rivers over twenty years of daily records. A simple two-component model explains most of the variation: a slow annual cycle driven by snow melt and a fast response to individual rain events. The model is easy to fit and gives useful forecasts a few days ahead.

## 1 Introduction

Small rivers react quickly to rain, and their discharge is harder to forecast than that of large basins. Local authorities nevertheless need such forecasts to plan flood defences and water abstraction. Previous studies focused on large catchments where gauges are dense and records long.

In this paper we use the records of eleven gauges that have been operated by volunteers since 2006. The data are noisy but complete, and they cover catchments between twelve and ninety square kilometres. We show that a model with only four parameters per river reproduces the seasonal pattern and most of the short-term peaks.

## 2 Data

Each gauge reports the water level every fifteen minutes. Levels are converted to discharge with rating curves that are checked twice a year. We average the readings to daily values and remove days on which the gauge was frozen or under maintenance, which affects less than one percent of all days.

Rainfall comes from the nearest weather station, at most eight kilometres from the gauge. Snow depth is available for six of the eleven catchments; for the others we estimate it from temperature and precipitation with a standard degree-day method.

# Measurement of document conversion

Elena Martin | ID: DOC-109-1 | 2026-10-15

## 1 Introduction

The service begins on 15 October 2026 and continues for twelve months. Written approval is required before any additional work. The reference number must appear on every payment.

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The service begins on 15 October 2026 and continues for twelve months. Written approval is required before any additional work. The reference number must appear on every payment.

## 2 Results

F1 = 2PR / (P + R); n = 128; p = 0.032.

| Description | Qty | Unit price | Amount |
|---|---|---|---|
| Office paper | 1 | 12.35 | 12.35 |
| Delivery service | 2 | 15.55 | 31.10 |
| Maintenance kit | 3 | 18.75 | 56.25 |

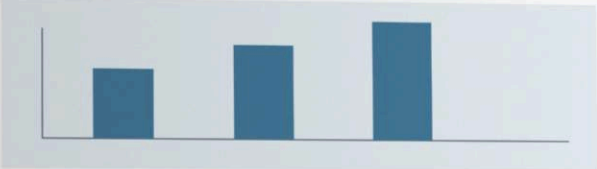


Figure 1. Quarterly observations

## References

[1] Martin, E. Document analysis. 2024. pp. 12-18.

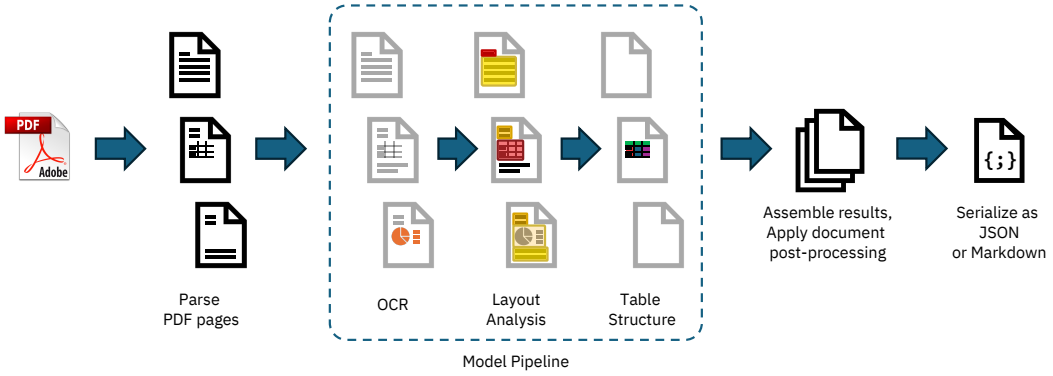



Figure 1: Sketch of Docling's default processing pipeline. The inner part of the model pipeline is easily customizable and extensible.

licensing (e.g. pymupdf [7]), poor speed or unrecoverable quality issues, such as merged text cells across far-apart text tokens or table columns (pypdfium, PyPDF) [15, 14].

We therefore decided to provide multiple backend choices, and additionally open-source a custom-built PDF parser, which is based on the low-level *qpdf* [4] library. It is made available in a separate package named *docling-parse* and powers the default PDF backend in Docling. As an alternative, we provide a PDF backend relying on *pypdfium*, which may be a safe backup choice in certain cases, e.g. if issues are seen with particular font encodings.

### 3.2 AI models

As part of Docling, we initially release two highly capable AI models to the open-source community, which have been developed and published recently by our team. The first model is a layout analysis model, an accurate object-detector for page elements [13]. The second model is TableFormer [12, 9], a state-of-the-art table structure recognition model. We provide the pre-trained weights (hosted on huggingface) and a separate package for the inference code as *docling-ibm-models*. Both models are also powering the open-access deepsearch-experience, our cloud-native service for knowledge exploration tasks.

#### Layout Analysis Model

Our layout analysis model is an object-detector which predicts the bounding-boxes and classes of various elements on the image of a given page. Its architecture is derived from RT-DETR [16] and re-trained on DocLayNet [13], our popular human-annotated dataset for document-layout analysis, among other proprietary datasets. For inference, our implementation relies on the *onnxruntime* [5].

The Docling pipeline feeds page images at 72 dpi resolution, which can be processed on a single CPU with sub-second latency. All predicted bounding-box proposals for document elements are post-processed to remove overlapping proposals based on confidence and size, and then intersected with the text tokens in the PDF to group them into meaningful and complete units such as paragraphs, section titles, list items, captions, figures or tables.

#### Table Structure Recognition

The TableFormer model [12], first published in 2022 and since refined with a custom structure token language [9], is a vision-transformer model for table structure recovery. It can predict the logical row and column structure of a given table based on an input image, and determine which table cells belong to column headers, row headers or the table body. Compared to earlier approaches, TableFormer handles many characteristics of tables, such as partial or no borderlines, empty cells, rows or columns, cell spans and hierarchy both on column-heading or row-heading level, tables with inconsistent indentation or alignment and other complexities. For inference, our implementation relies on *PyTorch* [2].

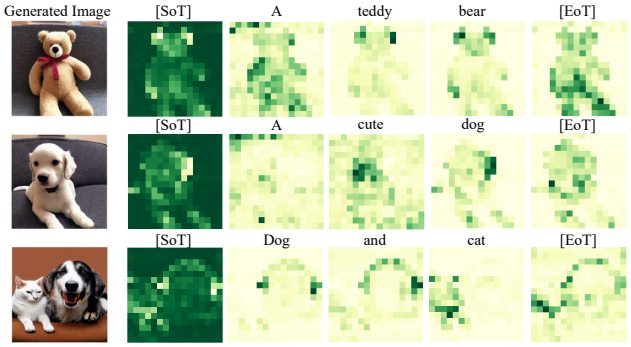



Figure 3: Cross-attention maps of different text prompts at the generation process. All of the examples indicate that start token ( [SoT] ) and padding tokens ( [EoT] ) carry rich semantic and layout information.

There is one more aspect of the SD architecture design that we require for our purposes. While there are several important design choices that make the network $D$ work well in practice, the mechanism that is of interest in our investigation is *cross-attention*, which allows the generation process to be conditioned on text, thus connecting visual and textual information. Each cross-attention layer takes an intermediate feature tensor $\boldsymbol{z}^{(\gamma)} \in \mathbb{R}^{C \times \frac{H}{r} \times \frac{W}{r}}$ as input, where $\gamma$ is the index of the relevant layer in the network, and $r$ is a scaling factor defining the spatial resolution at that level of the representation. The cross-attention map $A^{(\gamma)}$ associates each spatial location $u \in \{1, \dots, \frac{H}{r}\} \times \{1, \dots, \frac{W}{r}\}$ to a token indexed by $i \in \{1, \dots, N\}$:

$$A^{(\gamma)}_{ui} = \frac{\exp\langle \boldsymbol{z}^{(\gamma)}_u, K^{(\gamma)}_i \rangle}{\sum_{j=1}^{N} \exp\langle \boldsymbol{z}^{(\gamma)}_u, K^{(\gamma)}_j \rangle}, \quad \boldsymbol{a}^{(\gamma)}_u = \sum_{i=1}^{N} A^{(\gamma)}_{ui} V^{(\gamma)}_i,$$

where the value $V^{(\gamma)}_i$ and the key $K^{(\gamma)}_i$ are linear transformations of the token embedding $Y_i$ provided by the textual encoder, and $\boldsymbol{a}^{(\gamma)}_u$ is the output of the cross-attention layer.

## 3.2. Layout Guidance

Next, we introduce our method for training-free layout guidance. It has already been shown that cross-attention layers regulate the spatial layout of the generated image [12]. Specifically, the attention map the attention map $A^{(\gamma)}_{ui}$ determines how strongly each location $u$ in layer $\gamma$ is associated with each of the $N$ text tokens $y_i$. Since the sum of association strengths $\sum_{i=1}^{N} A^{(\gamma)}_{ui} = 1$ for each spatial location $u$, the different tokens can be thought of as "competing" for association with that location. To control the image layout using a bounding box B that corresponds to token $y_i$, the attention can be biased such that locations $u \in B$ within the target box are strongly associated with the token $y_i$ (while other locations are not). We introduce two ways of

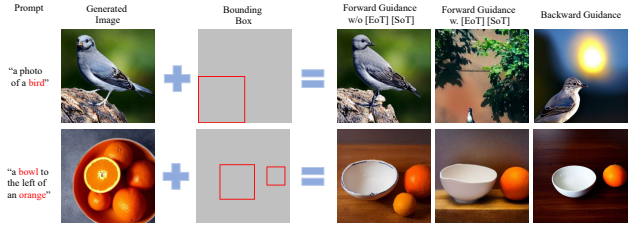



Figure 4: Comparison between forward and backward guidance, including guidance of start and padding tokens.

achieving this effect, which we refer to as forward guidance (Section 3.2.1) and backward guidance (Section 3.2.2). We give an overview of the two strategies in Figure 2.

### 3.2.1 Forward Guidance

In forward guidance, we represent the bounding box $B$ as a smooth windowing function $g^{(\gamma)}_u$ which is equal to a constant $c > 0$ inside the box and quickly falls to zero outside.[1] We rescale the windowing function to sum to one ($\|g^{(\gamma)}\|_1 = 1$). Then, we bias the attention map $A^{(\gamma)}$ by replacing it via:

$$A^{(\gamma)}_{ui} \leftarrow (1 - \lambda) A^{(\gamma)}_{ui} + \lambda g^{(\gamma)}_u \sum_v A^{(\gamma)}_{vi}, \tag{1}$$

where $\lambda \in [0, 1]$ defines the strength of the intervention. Note that: (1) only the cross-attention map $A^{(\gamma)}_{ui}$ corresponding to token $y_i$ is affected by this intervention (2) the window is weighed by the mass $\sum_v A^{(\gamma)}_{vi}$ so as to leave the latter unchanged.

This intervention is injected at each application of the denoiser function $D$ at a selected subset of layers $\gamma \in \Gamma$. This means that the activation maps computed by each of the selected layers are replaced with Equation (1).

### 3.2.2 Backward Guidance

The second approach we consider is backward guidance. In this case, we bias the attention $A^{(\gamma)}_{ui}$ by introducing the energy function

$$E(A^{(\gamma)}, B, i) = \left(1 - \frac{\sum_{u \in B} A^{(\gamma)}_{ui}}{\sum_u A^{(\gamma)}_{ui}}\right)^2 \tag{2}$$

This is used in a manner similar to *classifier guidance* [3]. Specifically, at each application of the denoiser $D$, when layer $\gamma \in \Gamma$ is evaluated, the gradient of the loss (2) is computed via backpropagation to update the latent $\boldsymbol{z}_t (\equiv \boldsymbol{z}^{(0)}_t)$:

$$\boldsymbol{z}_t \leftarrow \boldsymbol{z}_t - \sigma_t \eta \nabla_{\boldsymbol{z}_t} \sum_{\gamma \in \Gamma} E(A^{(\gamma)}, B, i), \tag{3}$$

[1] For simplicity, in our implementation, we put a Gaussian blob with $\sigma$ decided by the resolution, height, and width of the bounding box.




# The NIST Cybersecurity Framework (CSF) 2.0

National Institute of Standards and Technology
This publication is available free of charge from: https://doi.org/10.6028/NIST.CSWP.29

February 26, 2024

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
U.S. DEPARTMENT OF COMMERCE

## 3. Introduction to CSF Profiles and Tiers

This section defines the concepts of CSF Profiles and Tiers.

### 3.1. CSF Profiles

A *CSF Organizational Profile* describes an organization's current and/or target cybersecurity posture in terms of the Core's outcomes. Organizational Profiles are used to understand, tailor, assess, prioritize, and communicate the Core's outcomes by considering an organization's mission objectives, stakeholder expectations, threat landscape, and requirements. An organization can then prioritize its actions to achieve specific outcomes and communicate that information to stakeholders.

Every Organizational Profile includes one or both of the following:

1. A *Current Profile* specifies the Core outcomes that an organization is currently achieving (or attempting to achieve) and characterizes how or to what extent each outcome is being achieved.
2. A *Target Profile* specifies the desired outcomes that an organization has selected and prioritized for achieving its cybersecurity risk management objectives. A Target Profile considers anticipated changes to the organization's cybersecurity posture, such as new requirements, new technology adoption, and threat intelligence trends.

> A *Community Profile* is a baseline of CSF outcomes that is created and published to address shared interests and goals among a number of organizations. A Community Profile is typically developed for a particular sector, subsector, technology, threat type, or other use case. An organization can use a Community Profile as the basis for its own Target Profile. Examples of Community Profiles can be found on the NIST CSF website.

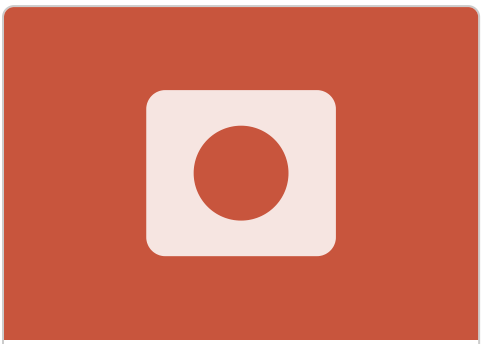
The steps shown in Fig. 3 and summarized below illustrate one way that an organization could use an Organizational Profile to help inform continuous improvement of its cybersecurity.



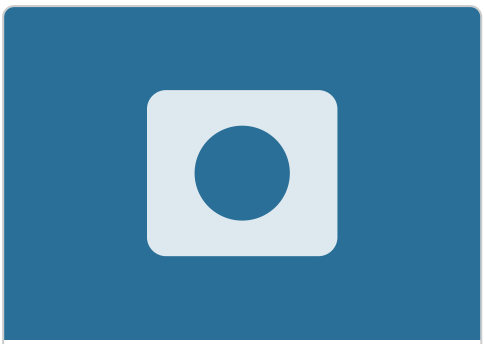

**Fig. 3. Steps for creating and using a CSF Organizational Profile**

# Summit Outfitters — Autumn Catalog

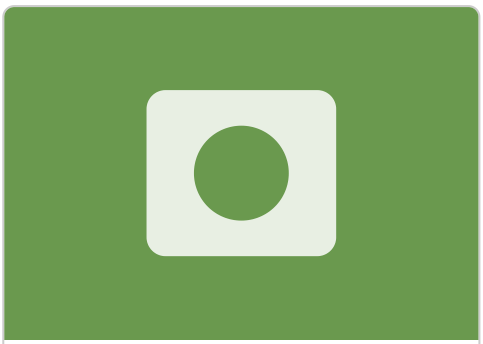
Prices in USD · free shipping over $75 · page 12



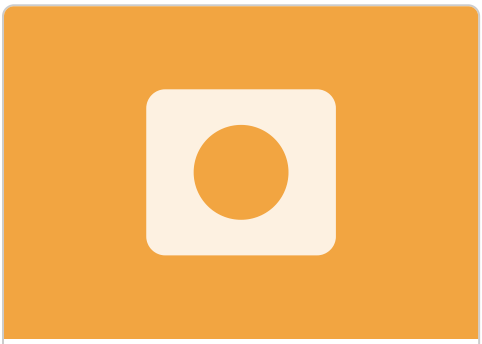
**Trail backpack 28 L**
SKU BP-2801
**$89.00**



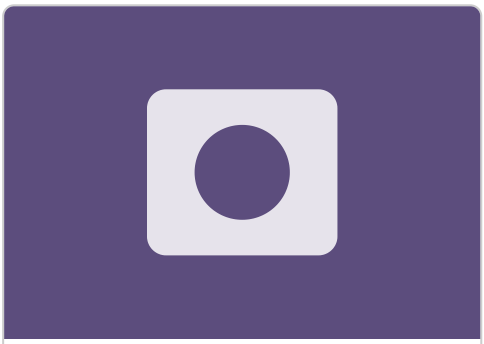
**Insulated bottle 750 ml**
SKU IB-0750
**$24.50**



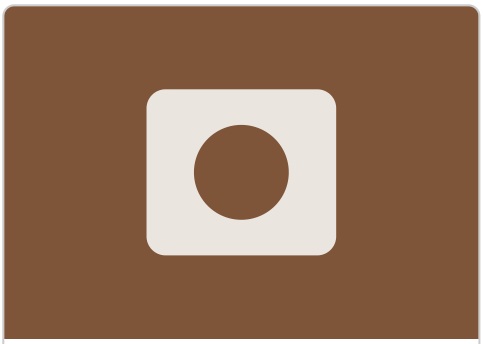
**Merino socks (2-pack)**
SKU MS-0202
**$18.00**



**Headlamp 400 lm**
SKU HL-0400
**$39.90**



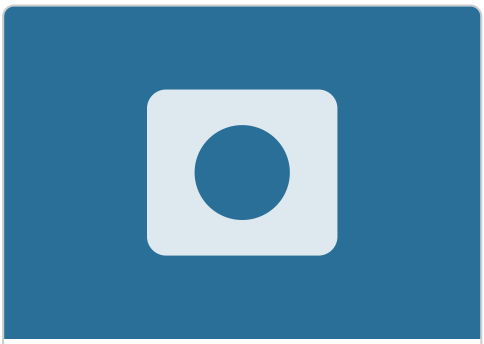
**Packable rain shell**
SKU RS-1100
**$129.00**



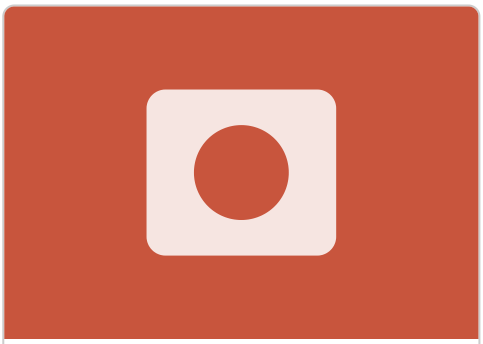
**Camp stove, compact**
SKU CS-0090
**$54.00**



**Merino socks (2-pack) – limited**
SKU MS-0202-L
**$15.30**



**Insulated bottle 750 ml – limited**
SKU IB-0750-L
**$20.82**



**Trail backpack 28 L – limited**
SKU BP-2801-L
**$75.65**

# Mercy Valley Clinical Laboratory

Complete blood count and metabolic panel — final report

| PATIENT | DATE OF BIRTH | PATIENT ID | SEX |
|---|---|---|---|
| Sample, Jordan A. | 1984-03-17 | MV-0048213 | F |
| COLLECTED | RECEIVED | REPORTED | ORDERING PHYSICIAN |
| 2026-11-09 08:15 | 2026-11-09 10:02 | 2026-11-10 07:40 | Dr. R. Okafor |

| Test | Result | Flag | Units | Reference range |
|---|---|---|---|---|
| **HEMATOLOGY** | | | | |
| White blood cell count | 11.8 | H | 10^9/L | 4.0 – 11.0 |
| Red blood cell count | 4.62 | | 10^12/L | 3.90 – 5.10 |
| Hemoglobin | 11.2 | L | g/dL | 12.0 – 15.5 |
| Hematocrit | 35.1 | L | % | 36.0 – 46.0 |
| Platelet count | 248 | | 10^9/L | 150 – 400 |
| **CHEMISTRY** | | | | |
| Glucose, fasting | 101 | H | mg/dL | 70 – 99 |
| Creatinine | 0.82 | | mg/dL | 0.50 – 1.10 |
| Sodium | 139 | | mmol/L | 135 – 145 |
| Potassium | 3.4 | L | mmol/L | 3.5 – 5.1 |
| Total cholesterol | 212 | H | mg/dL | < 200 |
| HDL cholesterol | 58 | | mg/dL | > 50 |

H = above reference range; L = below reference range. Results relate only to the sample received. Fasting status reported by patient. Reference ranges are adult values.

arXiv:2304.03373v1 [cs.CV] 6 Apr 2023

# Training-Free Layout Control with Cross-Attention Guidance

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Visual Geometry Group, University of Oxford
{minghao, iro, vedaldi}@robots.ox.ac.uk
silent-chen.github.io/layout-guidance

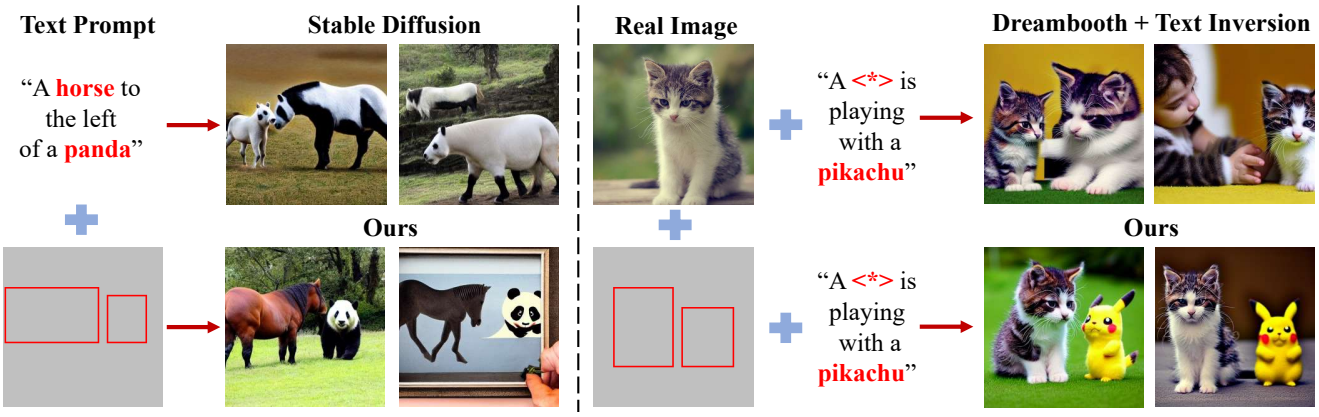



Figure 1: Left: our method can control the layout of an image generated by a pre-trained diffusion model, such as Stable Diffusion [31], without any finetuning or training. It also alleviates the problem that Stable Diffusion may not generate all objects in the prompt. Right: given a single real image, we manage to edit its position, context with the Dreambooth [33], and Text Inversion [8] technique. $\langle * \rangle$ is the learned token that encodes the object in the image.

## Abstract

*Recent diffusion-based generators can produce high-quality images based only on textual prompts. However, they do not correctly interpret instructions that specify the spatial layout of the composition. We propose a simple approach that can achieve robust layout control without requiring training or fine-tuning the image generator. Our technique, which we call layout guidance, manipulates the cross-attention layers that the model uses to interface textual and visual information and steers the reconstruction in the desired direction given, e.g., a user-specified layout. In order to determine how to best guide attention, we study the role of different attention maps when generating images and experiment with two alternative strategies, forward and backward guidance. We evaluate our method quantitatively and qualitatively with several experiments, validating its effectiveness. We further demonstrate its versatility by extending layout guidance to the task of editing the layout and context of a given real image.*

## 1. Introduction

Generative AI is easily one of the most disruptive technologies that have emerged in the last few years. In computer vision, new text-conditioned image generation methods, such as DALL-E [29], Imagen [34], and Stable Diffusion [31], have demonstrated that machines can generate image outputs of a quality sufficient to use in many applications. These methods can easily multiply the productivity of professional artists as well as laypeople.

Despite these successes, however, many practical applications of image generation, particularly in a professional context, require a high level of *control* that current methods lack. Specifications in language-based image generators are textual; and while text can tap into a vast library of high-level concepts, it is a poor vehicle for expressing fine-grained visual details in an image. In particular, text is a poor way of describing the precise *layout of a composition*.

In fact, as shown in previous works [9] and as we show in our experiments, state-of-the-art image generators are not even very good at correctly interpreting simple layout in-